

Module 2

Machine Learning Fundamentals I

2.1 Machine Learning Definition

- Machine Learning(ML) is a subfield of Artificial Intelligence (AI).
- A ML algorithm is an algorithm that is able to learn from data.
- Note that deep learning is a sub-field of ML.

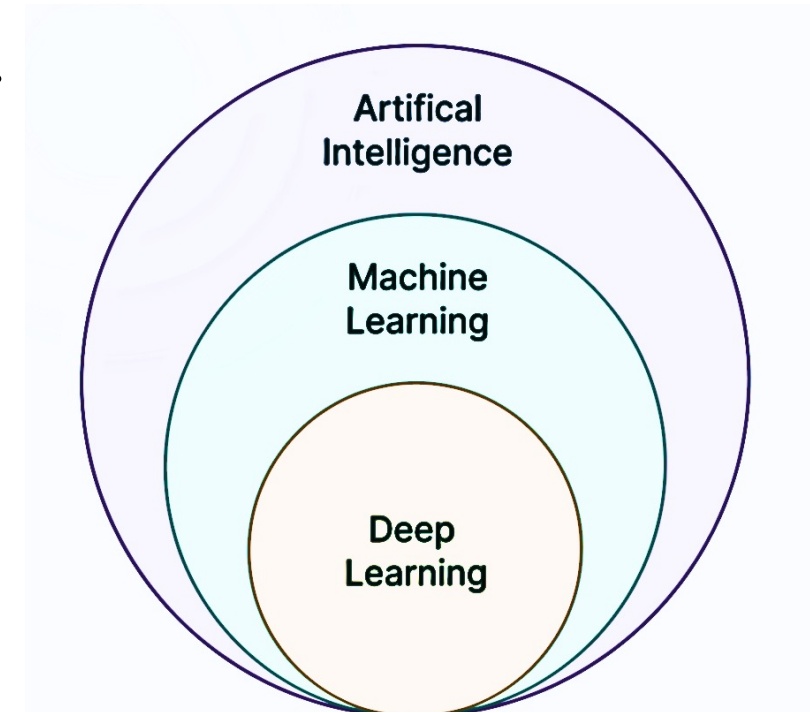


Fig. 2.1: Machine Learning as a Subfield of Artificial Intelligence

- Formally, **Arthur Samuel(1959)** defined ML as “the field of study that gives computers the ability to *learn* without being explicitly programmed.”

The 3 broad categories of ML are:

- i) Supervised Learning
- ii) Unsupervised Learning
- iii) Reinforcement Learning

- Details on Supervised ML will be presented later in this course.



Arthur Samuel

2.1.1 Machine Learning Tasks

- With ML, we can tackle tasks in T that are difficult to solve with explicit computer programs. For instance, if we build a chatbot to translate English to Yoruba, then *translation* is the *task*.
- ML **tasks** are usually described in terms of how the ML system should process an **example**.
- An **example** is a collection of **features** that have been quantitatively measured from some objects or events that we want the ML system to process.

- An **example** is formally represented as a vector:

$$\mathbf{x} \in \mathbb{R}^n \quad (2.1)$$

where each entry x_i of the vector is another feature.

- For instance, the features of a grayscale image will be the pixel values (0 to 255) in the image.

The following are common tasks in ML:

i) Classification Task

- In this type of task, the ML model/program is asked to specify which of k categories some input belongs. In order to solve the task, the learning algorithm is usually asked to produce a discrete valued-function:

$$f: \mathbb{R}^n \rightarrow \{1, \dots, k\} \quad (2.2)$$

- When $y = f(\mathbf{x})$, the ML model assigns an input described by vector $\mathbf{x}$ to a category identified by a discrete value y (also known as *class label*).

- Classification can be grouped into:
 - a) Binary classification:* distinguishes between two classes. One class can be seen as the positive (true) class and the other as the negative (false) class.
 - b) Multiclass classification:* distinguishes among more than two classes.
- An example of a classification task is object recognition in which the input is an image and the output is a numeric code, which is mapped to a user friendly class label (See Figures 2.2 and 2.3).

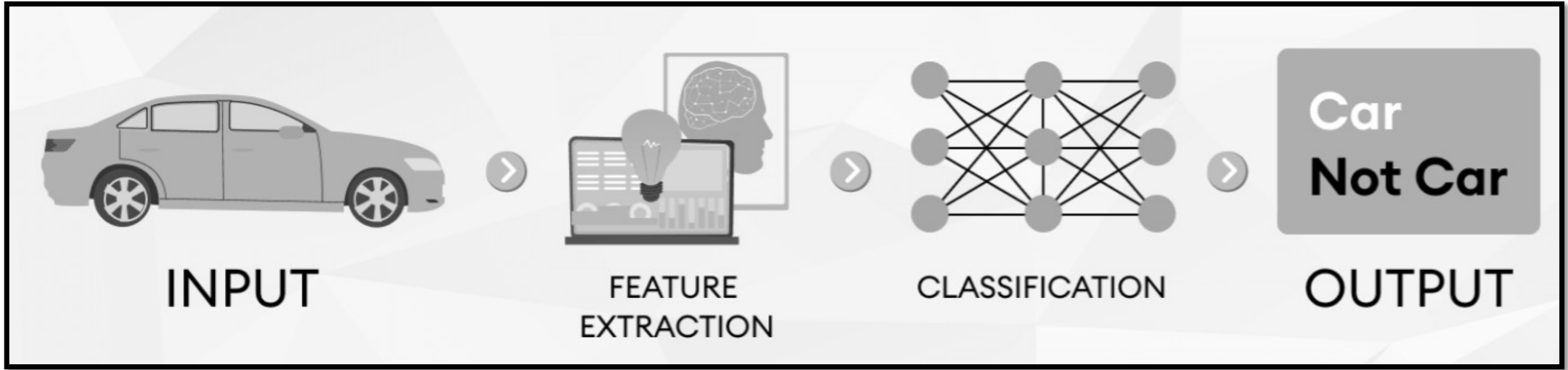


Fig. 2.2: Object Recognition with Shallow Machine Learning Model
(Source: <https://www.superannotate.com/blog/object-detection-with-deep-learning>)

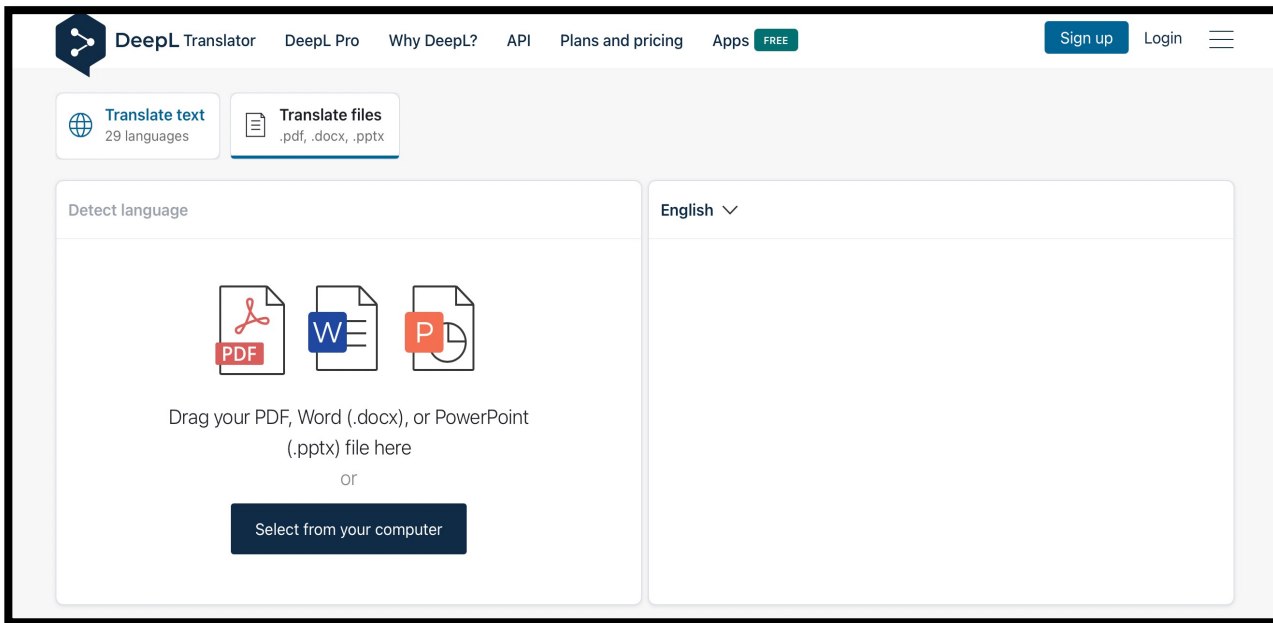
ii) Regression Task

- This task is similar to classification but the format of the output is different. In regression task, given some input, the ML model predicts a numerical value, which is continuous (or real). Thus, the learning algorithm is asked to produce a continuous-valued function:

$$f: \mathbb{R}^n \rightarrow \mathbb{R} \quad (2.3)$$

iii) Transcription Task

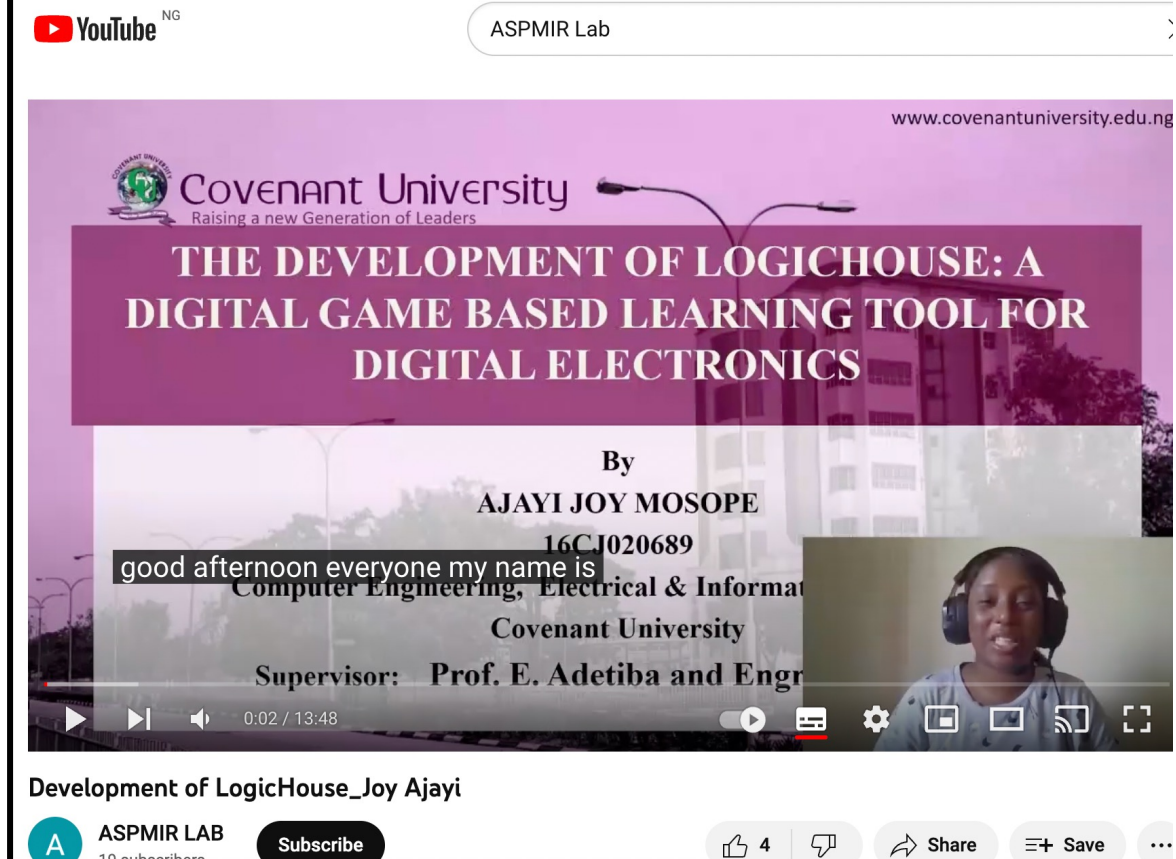
- In this task, the ML model transcribes relatively unstructured data into discrete textual form. ***Optical character recognition*** is an example of this task in which an image containing text is transcribed into sequence of characters or ***speech recognition*** in which an audio waveform is transcribed to sequences of the corresponding text.



<https://www.deepl.com/translator/files>

Fig. 2.3: Optical Character Recognition Sample

(Click the above link to use DeepL for OCR and other ML tasks)



<https://www.youtube.com/watch?v=Sck-JMhKtv8>

Fig. 2.4: Speech Recognition Sample

(Click the above link to see Google Speech Recognition in Action)

iv) Machine Translation Task

- This task involves the conversion of sequence of symbols in one language(e.g. English) to sequence of symbol in another language (e.g. French).

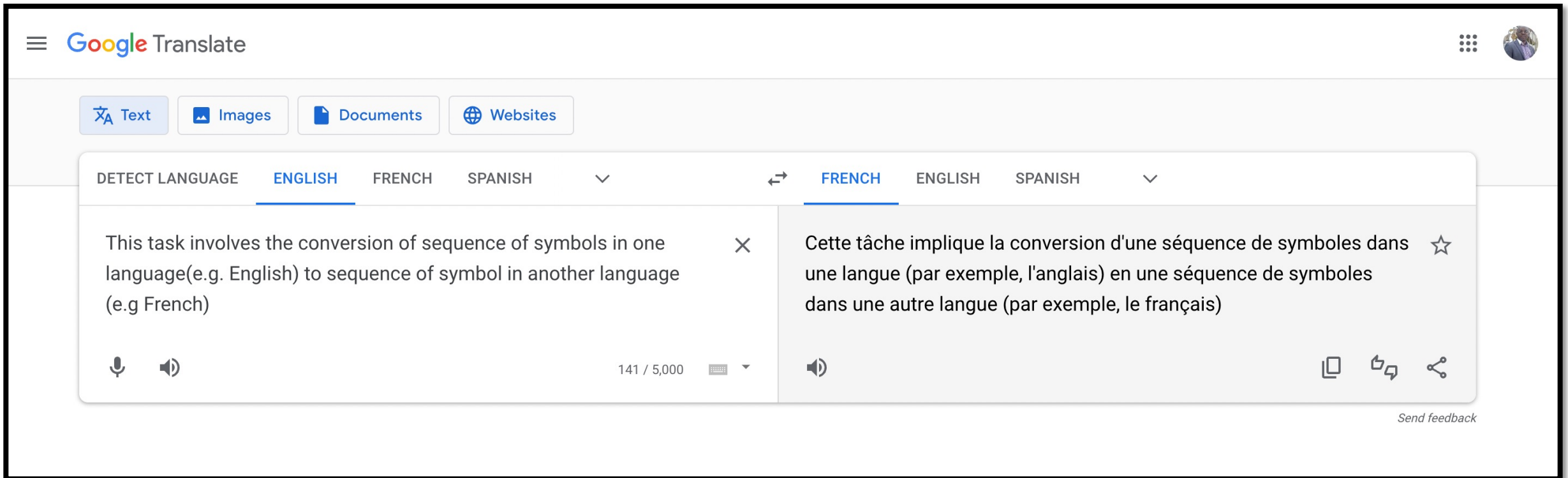


Figure 2.5: Machine Translation with Google Translate

v) *Synthesis Task*

- In this task, the ML model generates new examples that are similar to those in the training data. Examples of the tasks are as follows:

a) *Image or Video Synthesis*: **Deepfake** is an example of image or video synthesis task. **Deepfakes** are synthetic media in which an existing image or video of a human is replaced with someone else's likeness.

- Check (<https://deepfakesweb.com/>) for a web app for creating Deepfakes.

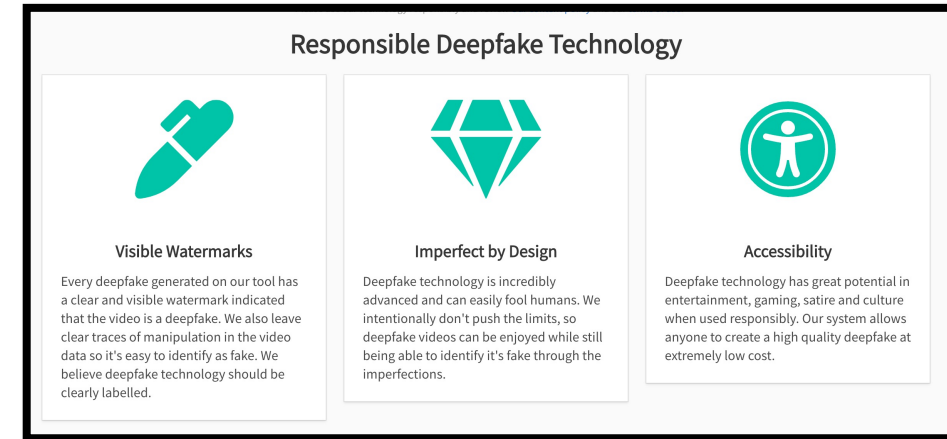
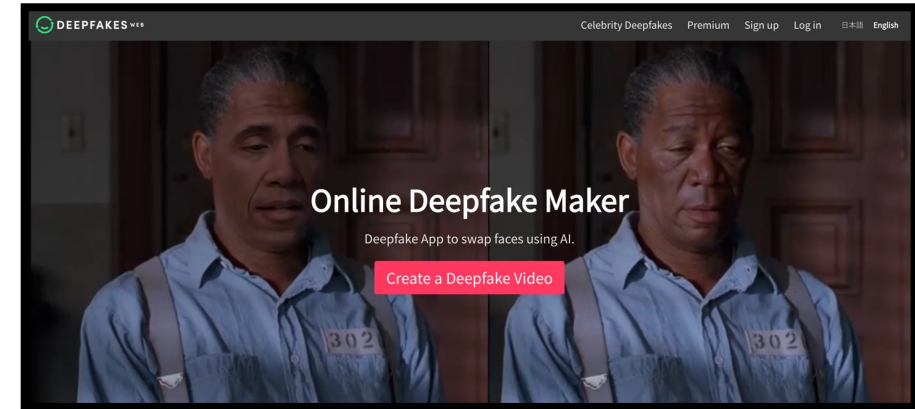


Figure 2.6: Deepfake WebApp Interface

b) Speech Synthesis: In this task, a written sentence is provided and the ML model produces an audio waveform that contains the spoken version of the sentence. the desire is for the sampling or synthesis procedure to generate a specific kind of output given the input. The **Google Translate App** in Fig. 2.5 contains speech synthesis capability.

c) Text-to-Image Synthesis: This task involves the conversion of text to the corresponding image of the textual meaning. In 2021, OpenAI released the **DALL-E** for text-to-image synthesis tasks.

- The Images on the right were generated at PromptBase (<https://promptbase.com/generate>), a web platform that is based on DALL-E AI model.
- The text prompts that synthesized the corresponding images are:

(a) Nelson Mandela speaking on a podium.

(b) Desmond Tutu speaking on a podium.

(c) Bishop David Oyedepo speaking on a podium.

(d) Pst E. A. Adeboye speaking on a podium.



a



b



c



d

Figure 2.7: Text-to-Image Synthesis

(<https://promptbase.com/generate>)

(Click the above link to use Promptbase for text-to-image tasks)

2.1.2 Overview of Performance Measure in ML

- Performance Measure allows us to evaluate how well the ML algorithm performs on data that it has not seen before since this determines how well it will work when deployed in the real world. Thus a **test set** of data that is different from the **training set**, used to build the ML model is normally used.
- The following are two key performance measures in ML:
 - i) **Accuracy:** This is a measure of the proportion of examples for which the model produces the correct output. It is used for tasks such as classification and transcription.

ii) **Error rate:** Equivalent information to accuracy can be obtained by computing the error rate such as the **Mean Square Error (MSE)** for a classification or transcription task. It is the proportion of examples for which the model produces an incorrect output.

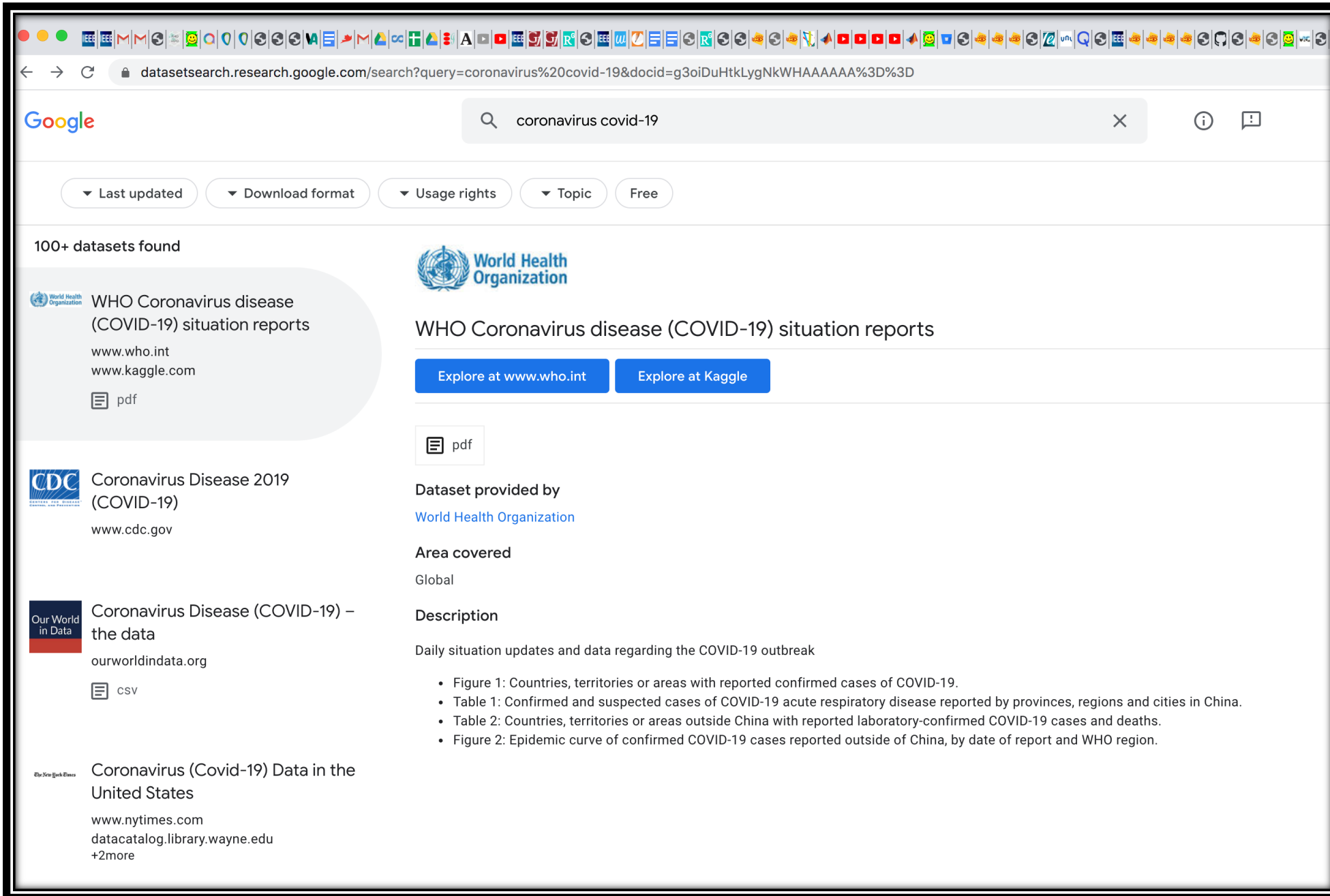
2.1.3 Datasets for Machine Learning

- One of the most important parts in building ML models is an understanding of the **dataset** you are working with and how it relates to the task you want to solve.
- Dataset collection and archiving are major components of ML model development for different tasks.
- Few of the repositories where datasets are archived for ML tasks are hereafter presented.

i) Google Dataset Search: This service was launched for searching repositories of datasets on the Web (<https://datasetsearch.research.google.com>).

❖ The service allows data providers to upload and describe their datasets using various metadata (e.g., author, publication date, how the data was collected, and terms for using the data) so that they become more searchable on line.

❖ Researchers can use the dataset, which comprises of annotated audio, image sets, and etc to check their hypotheses or train and test their machine learning models. See Fig. 3.8 for a web interface of the service.



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Figure 2.8: Google Dataset Search Web Interface

ii) IEEE DataPort

- ❖ It is an easily accessible data platform, which enables users to store, search, access and manage data (<https://ieee-dataport.org>).
- ❖ It accepts all formats and sizes of datasets up to 2TB and provides downloading capabilities for such datasets in the Cloud.

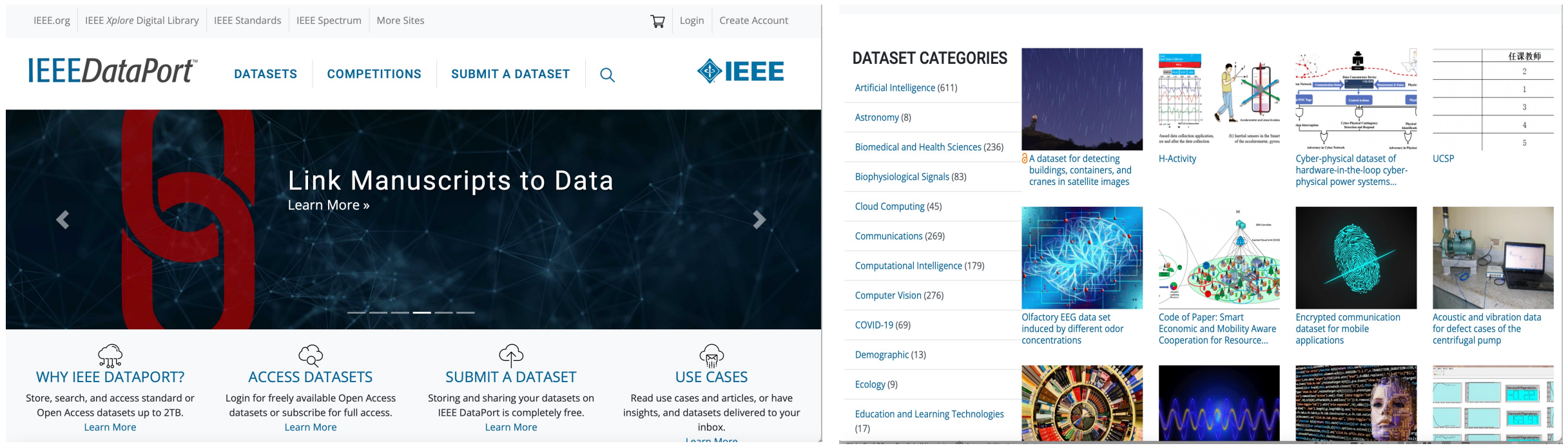


Figure 2.9: IEEE DataPort Web Interface

iii) UCI Machine Learning Repository

- ❖ University of California Irvine (UCI) Machine Learning Repository is a collection of datasets that are used by the ML community for empirical analysis of AI/ML algorithms (<https://archive.ics.uci.edu/ml/index.php>).
- ❖ The repository was created as an ftp archive in 1987 by David Aha and fellow graduate students at UC Irvine.
- ❖ It is used by students, educators, and researchers globally as a primary source of machine learning datasets.




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 Check out the [beta version](#) of the new UCI Machine Learning Repository we are currently testing! [Contact us](#) if you have any issues, questions, or concerns. [Click here to try out the new site.](#)

Browse Through:

Default Task
[Classification \(466\)](#)
[Regression \(151\)](#)
[Clustering \(121\)](#)
[Other \(56\)](#)

Attribute Type
[Categorical \(38\)](#)
[Numerical \(422\)](#)
[Mixed \(55\)](#)

Data Type
[Multivariate \(480\)](#)
[Univariate \(30\)](#)
[Sequential \(59\)](#)
[Time-Series \(126\)](#)
[Text \(69\)](#)
[Domain-Theory \(23\)](#)
[Other \(21\)](#)

Area
[Life Sciences \(147\)](#)
[Physical Sciences \(57\)](#)
[CS / Engineering \(234\)](#)
[Social Sciences \(41\)](#)
[Business \(45\)](#)
[Game \(12\)](#)
[Other \(84\)](#)

622 Data Sets

Table View
 [List View](#)

Name	Data Types	Default Task	Attribute Types	# Instances	# Attributes	Year
 Abalone	Multivariate	Classification	Categorical, Integer, Real	4177	8	1995
 Adult	Multivariate	Classification	Categorical, Integer	48842	14	1996
 Annealing	Multivariate	Classification	Categorical, Integer, Real	798	38	
 Anonymous Microsoft Web Data		Recommender-Systems	Categorical	37711	294	1998
 Arrhythmia	Multivariate	Classification	Categorical, Integer, Real	452	279	1998
 Artificial Characters	Multivariate	Classification	Categorical, Integer, Real	6000	7	1992

Figure 2.10: UCI Machine Learning Repository Web Interface

iv) Kaggle Datasets

- ❖ Kaggle is a subsidiary of Google since 2017. It is a platform where AI/ML enthusiasts find and publish datasets as well as build models in a web-based environment.
- ❖ It provides a platform to share codes with other AI and ML engineers/researchers and to enter competitions to solve AI challenges.
- ❖ In June 2017, Kaggle passed 1 million registered users, and as of 2021, it has over 8 million registered users.

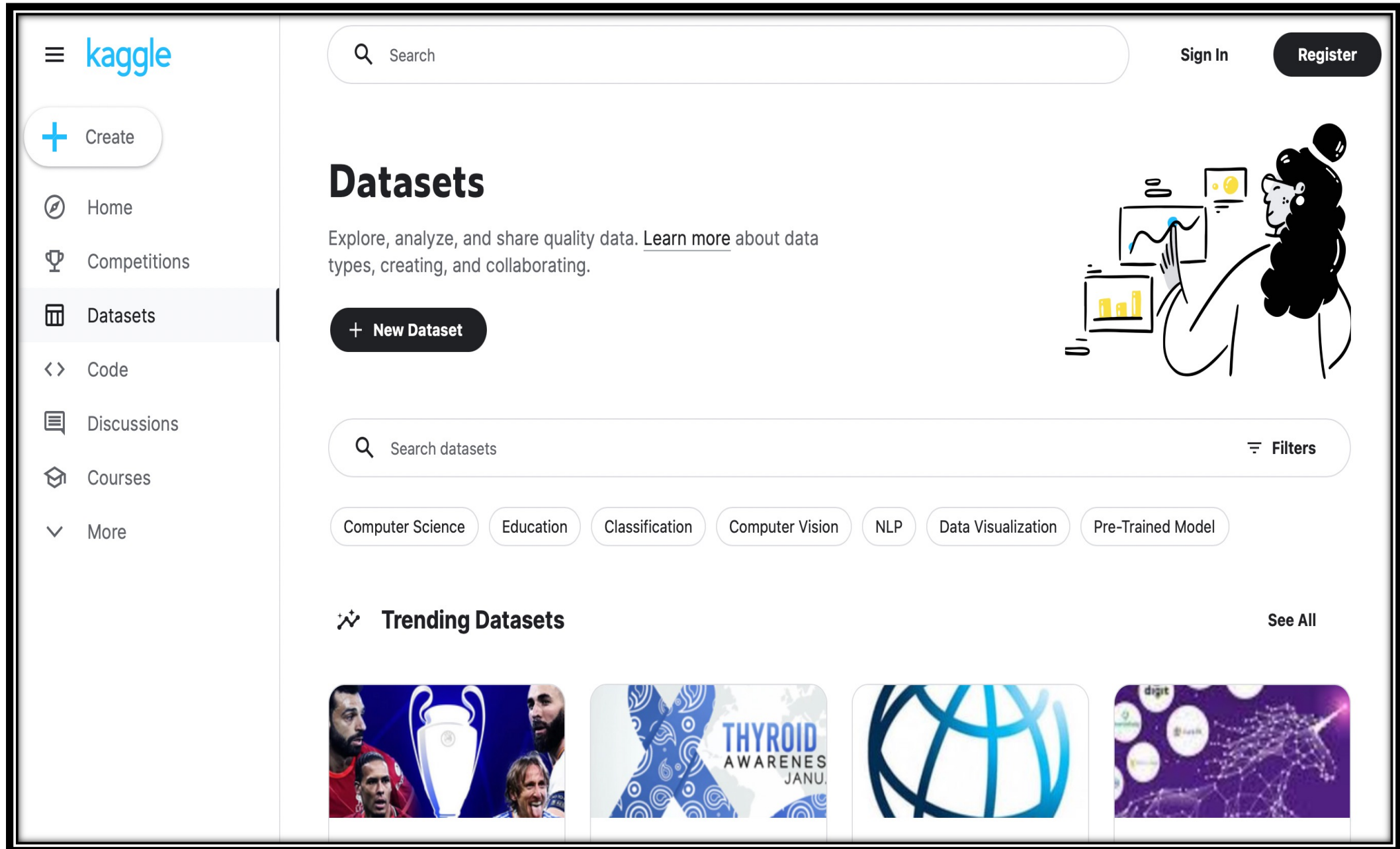


Fig. 2.11: Kaggle Datasets Web Interface

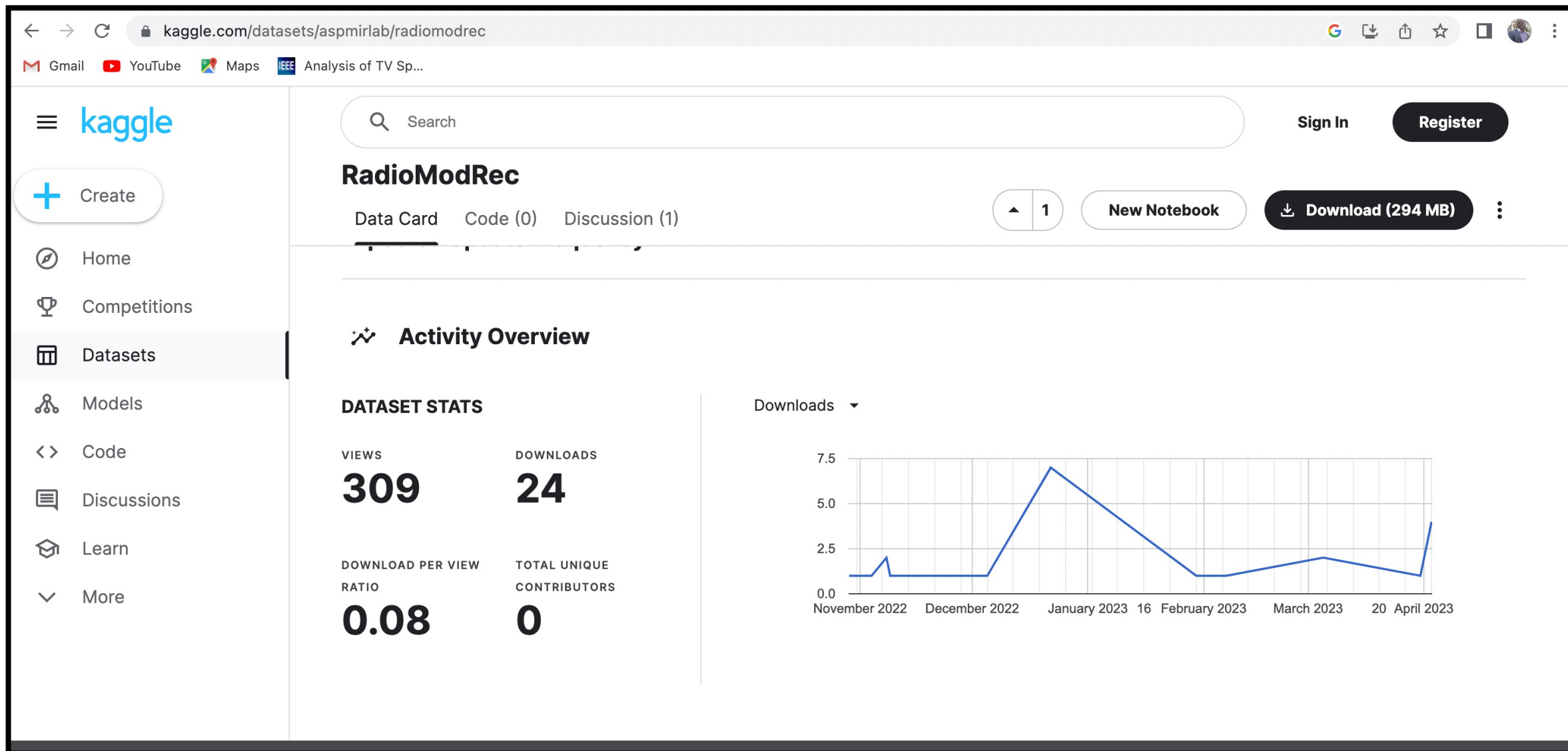


Fig. 2.12: RadioModRec Dataset by ASPMIR Lab on Kaggle

2.1.4 Computational Tools and Libraries for Machine Learning

- The algorithms and data in ML are only one part of a bigger process to implement a given task.
- Different programming languages, libraries, frameworks and computational tools are normally utilized to solve ML based problems.
- Python is one of the forefront languages for ML development works.
- It is selected for this course because it combines the power of general-purpose programming languages with the ease of use and rich toolboxes in domain-specific scripting languages such as MATLAB and R.
- Python has libraries for *data loading, visualization, statistics, image processing, creation of complex GUIs, web services* and integration into *embedded systems*.

2.1.4.1 Some Interactive Development Environments for ML

- i) **Jupyter Notebook:** It is an interactive environment for running code in the browser. It supports many programming languages but the choice language for this course is Python. It makes it easy to incorporate code, text, visualisations and images.

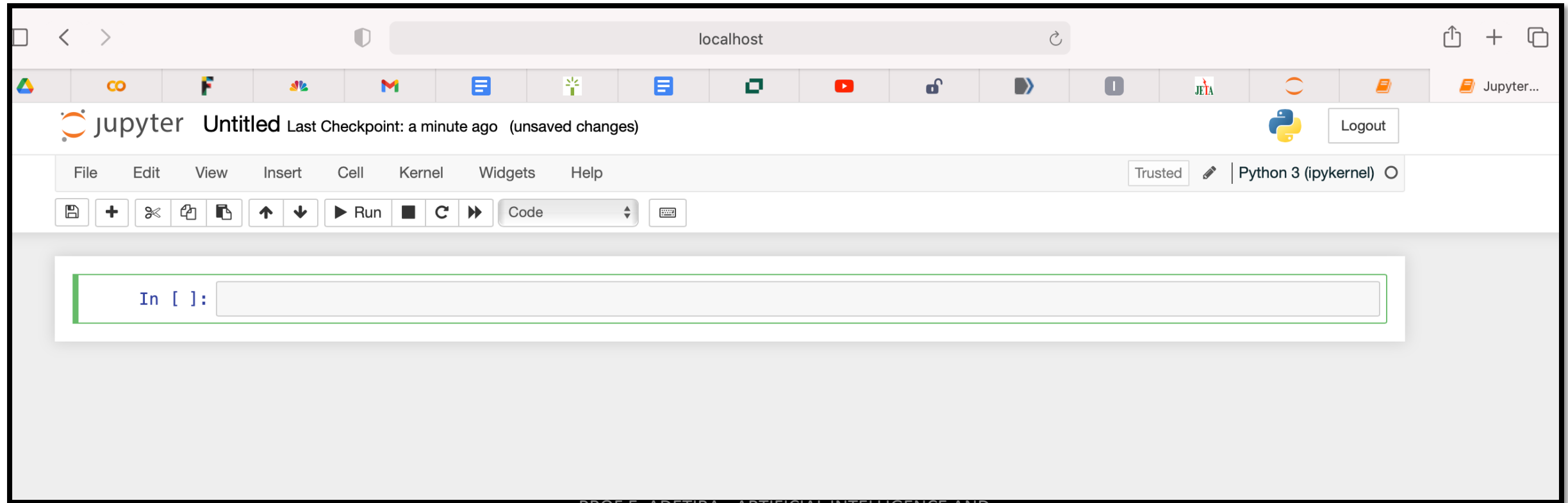


Fig. 2.13: Jupyter Notebook Interface from Anaconda Navigator

ii) Google Colab: Colab is a short form for Colaboratory. It is a hosted Jupyter notebook cloud service from Google Research. It allows programming in Python through the browser without the need for any setup by the user. It is a good platform for ML and collaborative development with free access to computing resources. Colab notebooks are stored in your Google Drive and you can share the notebooks or push them to GitHub. Colab Pro is an extension of Colab with paid extension for more computing resources such as RAMs, CPU Cores, GPUs and TPUs.

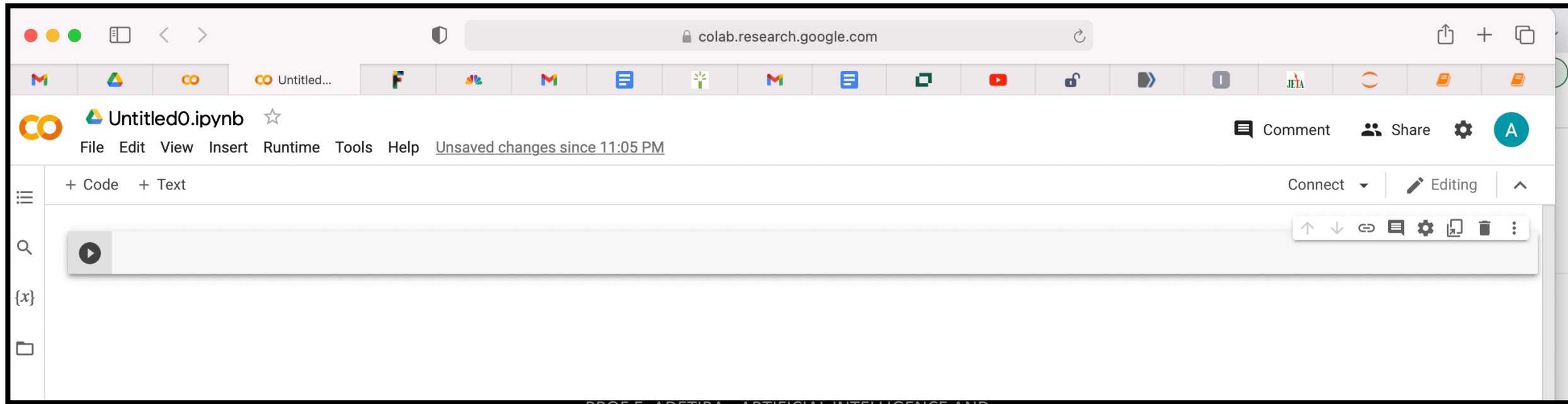


Fig. 2.14: Google Colab Interface for Cloud Based Interactive Coding

iii) **JupyterHub** JupyterHub brings the power of Jupyter notebooks to groups of users. It grants access to computational resources without the users having to bother about installation and maintenance.

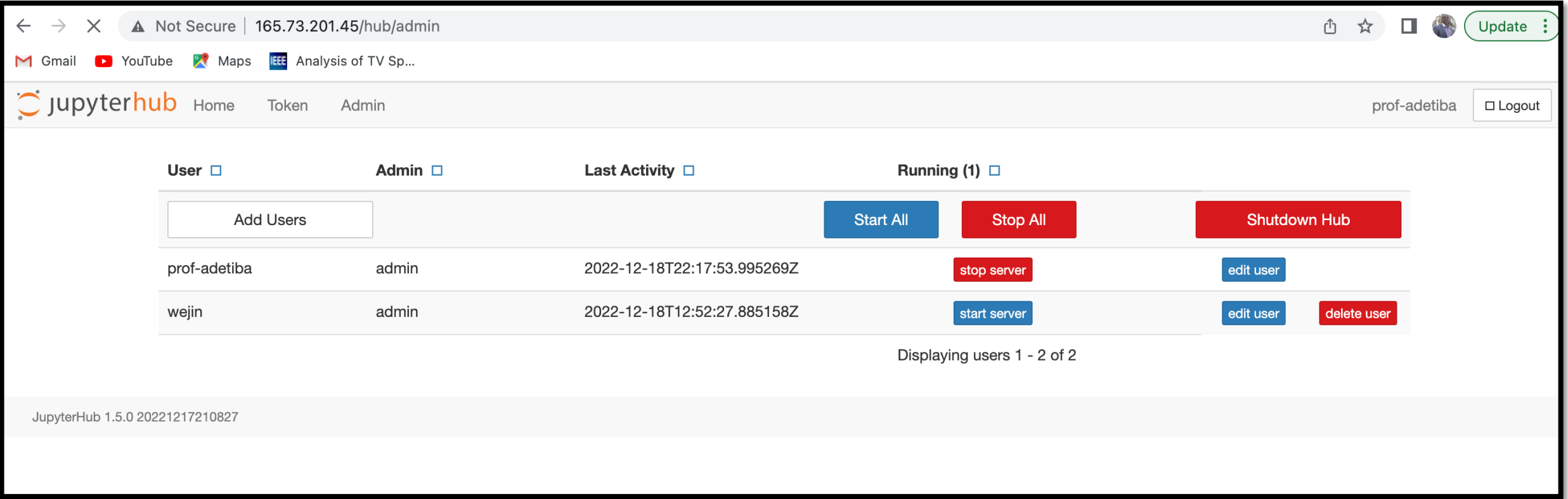


Fig. 2.15: A Sample JupyterHub Interface

2.1.4.2 Key Python Libraries for ML Programming Support

- i) **NumPy:** This is one of the primary libraries for scientific computing in Python. It contains functionality for multidimensional arrays. It also contains high-level mathematical functions such as linear algebra operations and Fourier transform.

Example 2.1: An example to demonstrate the use of NumPy

```
import numpy as np
# Create and print a 3 x 3 array
x = np.array([[1, 2, 3], [4, 5, 6], [7, 8, 9]])
print("x:\n",format(x))
print("")
# Create and print a 5 x 5 Identity Matrix using numpy array
eye = np.eye(5)
print("A 4x4 Identity Matrix:\n",format(eye))
```

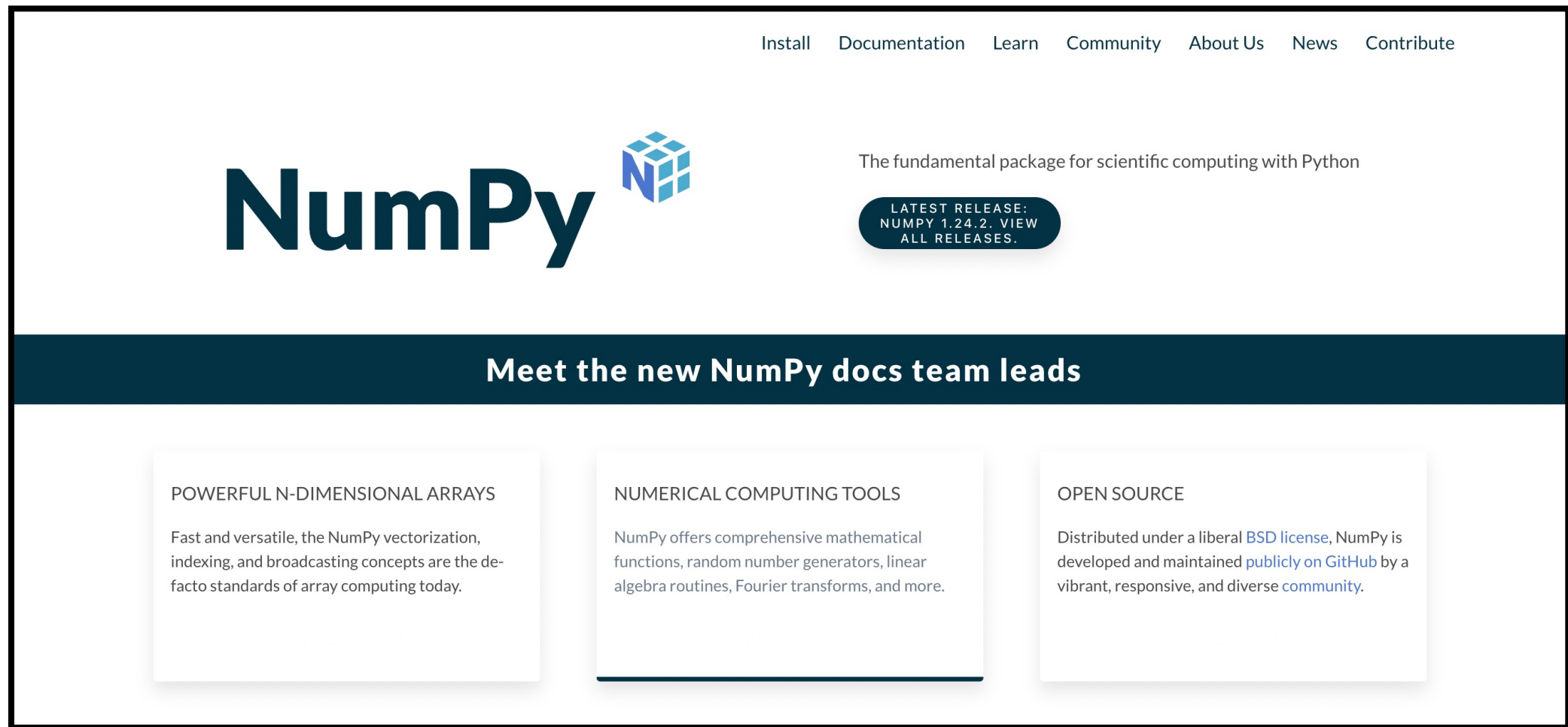


Fig. 2.16: Numpy User Interface
(<https://numpy.org>)

ii) Matplotlib: This is the primary scientific plotting library in Python. It contains functions for line charts, histograms, scatterplots and etc. You can show figures and other plots directly in Jupyter notebook using this library.

Example 2.2: An example to demonstrate the use of Matplotlib

```
import matplotlib.pyplot as plt  
#Generate a sequence of numbers from -10 to 10 with 500 steps  
y = np.linspace(-10,10,500)  
#Create a second array z as the cosine of y  
z = np.cos(y)  
#Use the plot function to plot array y against array z  
plt.plot(y,z,marker='x')
```

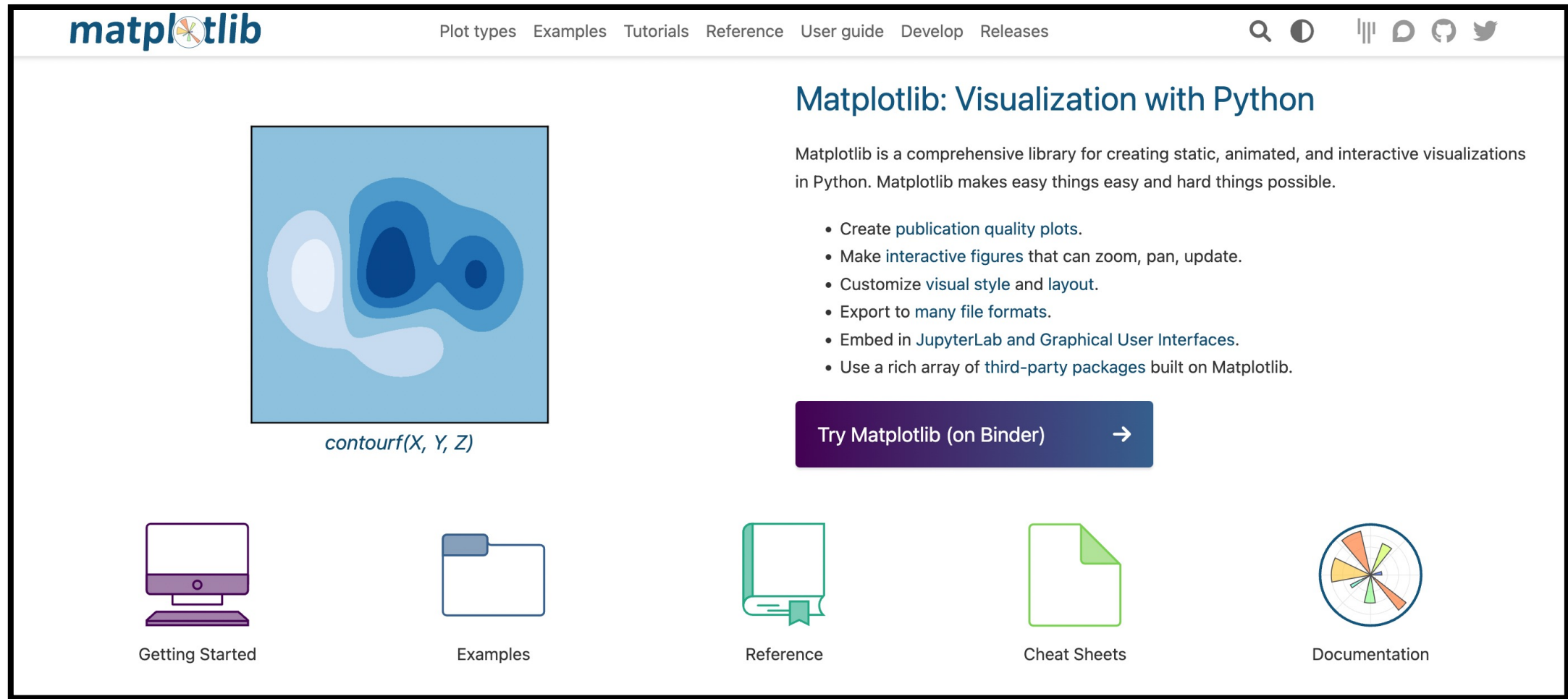


Fig. 2.17: Matplotlib User Interface
(<https://matplotlib.org>)

iii) Pandas: This is a Python library for data manipulation and analysis, which provides strong support for ML tasks.

- Pandas DataFrame is a table similar to Excel spreadsheet.
- Pandas provides different methods to operate on the table.
- It allows each column to have unique data type. It can also ingest data from varieties of file formats and databases such as Excel file, CSV and SQL.

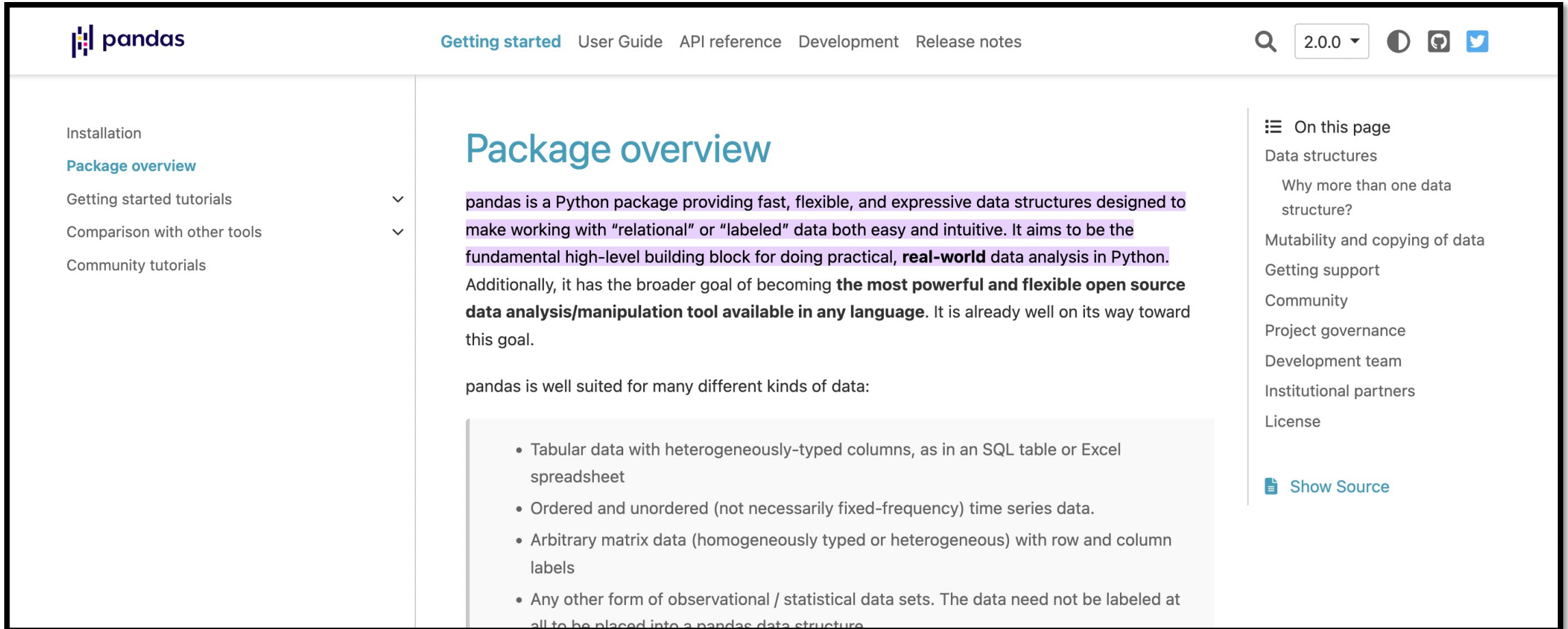


Fig. 2.18: Pandas User Interface
(<https://pandas.pydata.org>)

Example 2.3: An example to demonstrate the use of Pandas

```
import pandas as pd
from IPython.display import display

#Create a simple dataset of people using Dictionary type
data = {'Name': ["Segun", "Lawal", "Seyi", "Daniel"],
        'State': ["Ogun", "Imo", "Delta", "Kwara"],
        'Age': [18, 20, 26, 28]
        }

data_pandas = pd.DataFrame(data)

#IPython.display allows fine printing of dataframes in Jupyter
display(data_pandas)

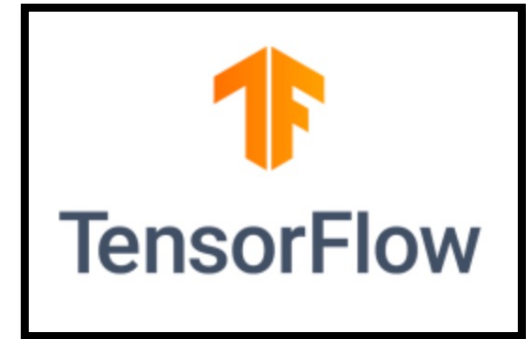
#Select all rows that have an age column less than or equal to 25
display(data_pandas[data_pandas.Age <= 25])
```

2.1.4.3 Key ML Libraries/Frameworks in Python

i) **Scikit-learn:** The scikit-learn is a very popular tool and one of the most prominent Python libraries for Machine Learning. It is an open source project, thus can be used and distributed freely by anyone. It works well with all the Python libraries earlier explained in the previous subsection. More on this later.



i) **TensorFlow:** It is a free framework that was developed by **Google Brain Team**. TensorFlow (TF) offers a powerful library, tools, and resources for numerical computation, specifically for large scale machine learning and deep learning projects. With TF, ML developers can build and deploy ML applications efficiently. It provides a high-level Keras API, which provide an easy start starters in ML. It has support for reinforcement learning, Computer Vision (CV), Natural Language Processing (NLP). TF enables you to run existing ML models using the *TensorFlow.js*.



TensorFlow Lite is a mobile library for deploying models on mobile, microcontrollers and other edge devices. TF can run on GPU for faster performance. ***TensforFlow Dataset (TFDS)*** is a collection of datasets for use with TF, and other ML frameworks.



iii) PyTorch: It is a free and an open-source ML framework developed by **FAIR(Facebook's AI Research)** lab. It has support for Python, C++ and Java, but the the Python interface is more interactive. It can also be used for Natural Language Processing (NLP), game development and computer vision tasks. It has support for GPU for fast execution of models.



Example 2.4: An example to demonstrate the use of TensorFlow Dataset (TFDS)

TensorFlow Dataset (TFDS) can be installed with:

```
!pip install -q tfds-nightly tensorflow
```

Import the required libraries

```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

```
import tensorflow as tf
```

```
import tensorflow_datasets as tfds
```

To get the list of available dataset, use tfds.list_builders()

tfds.list_builders()

##Load the dataset with tfds.load and convert the pandas.DataFrame with tfds.as_dataframe to Visualise the dataset

ds, info = tfds.load('mnist', split='train', shuffle_files=True, with_info=True)

tfds.as_dataframe(ds.take(4), info)

##tfds.show_examples returns a matplotlib.figure to show selected images

##(Note that only image datasets are supported now)

fig = tfds.show_examples(ds, info)

Module 2- Assignment

1) With relevant equations, describe the following performance measures in ML: *i) Accuracy ii) Precision iii) Recall iv) F-Measure*

2) Run Examples 2.1 to 2.4 in Jupyter Notebook

NOTE:

i) The two assignments **MUST** be saved (in a single notebook) as **fullname_matricnumber.ipynb**

ii) Your notebook **MUST** be submitted on MOODLE on or before the next class.